

Chapter 19

Definite Integrals

Only One Option Correct Type Questions

1. $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\int_{\frac{\pi}{4}}^x \sec^2 t \, dt}{x^2 - \frac{\pi^2}{16}}$ equals [IIT-JEE-2007 (Paper-1)]
- (A) $\frac{8}{\pi} f(2)$ (B) $\frac{2}{\pi} f(2)$
- (C) $\frac{2}{\pi} f\left(\frac{1}{2}\right)$ (D) $4f(2)$
2. Let f be a non-negative function defined on the interval $[0, 1]$. If $\int_0^x \sqrt{1 - (f'(t))^2} \, dt = \int_0^x f(t) \, dt$, $0 \leq x \leq 1$, and $f(0) = 0$, then [IIT-JEE-2009 (Paper-1)]
- (A) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$ (B) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$
- (C) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$ (D) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$
3. The value of $\int_{\sqrt{\ln 2}}^{\sqrt{\ln 3}} \frac{x \sin x^2}{\sin x^2 + \sin(\ln 6 - x^2)} \, dx$ is [IIT-JEE-2011 (Paper-1)]
- (A) $\frac{1}{4} \ln \frac{3}{2}$ (B) $\frac{1}{2} \ln \frac{3}{2}$
- (C) $\ln \frac{3}{2}$ (D) $\frac{1}{6} \ln \frac{3}{2}$
4. Let $f : [-1, 2] \rightarrow [0, \infty)$ be a continuous function such that $f(x) = f(1 - x)$ for all $x \in [-1, 2]$. Let $R_1 = \int_{-1}^2 x f(x) \, dx$, and R_2 be the area of the region bounded by $y = f(x)$, $x = -1$, $x = 2$, and the x -axis. Then [IIT-JEE-2011 (Paper-2)]
- (A) $R_1 = 2R_2$ (B) $R_1 = 3R_2$
- (C) $2R_1 = R_2$ (D) $3R_1 = R_2$

5. The value of the integral $\int_{-\pi/2}^{\pi/2} \left(x^2 + \ln \frac{\pi+x}{\pi-x} \right) \cos x dx$ is [IIT-JEE-2012 (Paper-2)]

- (A) 0 (B) $\frac{\pi^2}{2} - 4$
(C) $\frac{\pi^2}{2} + 4$ (D) $\frac{\pi^2}{2}$

6. The following integral $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2 \operatorname{cosec} x)^{17} dx$ is equal to [JEE (Adv)-2014 (Paper-2)]

- (A) $\int_0^{\log(1+\sqrt{2})} 2(e^u + e^{-u})^{16} du$ (B) $\int_0^{\log(1+\sqrt{2})} (e^u + e^{-u})^{17} du$
(C) $\int_0^{\log(1+\sqrt{2})} (e^u - e^{-u})^{17} du$ (D) $\int_0^{\log(1+\sqrt{2})} 2(e^u - e^{-u})^{16} du$

7. Let $f : [0, 2] \rightarrow \mathbb{R}$ be a function which is continuous on $[0, 2]$ and is differentiable on $(0, 2)$ with $f(0) = 1$. Let

$F(x) = \int_0^{x^2} f(\sqrt{t}) dt$ for $x \in [0, 2]$. If $F'(x) = f'(x)$ for all $x \in (0, 2)$, then $F(2)$ equals [JEE (Adv)-2014 (Paper-2)]

- (A) $e^2 - 1$ (B) $e^4 - 1$
(C) $e - 1$ (D) e^4

8. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^2 \cos x}{1+e^x} dx$ is equal to [JEE (Adv)-2016 (Paper-2)]

- (A) $\frac{\pi^2}{4} - 2$ (B) $\frac{\pi^2}{4} + 2$
(C) $\pi^2 - e^{\frac{\pi}{2}}$ (D) $\pi^2 + e^{\frac{\pi}{2}}$

One or More Option(s) Correct Type Questions

9. Let $S_n = \sum_{k=1}^n \frac{n}{n^2 + kn + k^2}$ and $T_n = \sum_{k=0}^{n-1} \frac{n}{n^2 + kn + k^2}$, for $n = 1, 2, 3, \dots$ then, [IIT-JEE-2008 (Paper-1)]

- (A) $S_n < \frac{\pi}{3\sqrt{3}}$ (B) $S_n > \frac{\pi}{3\sqrt{3}}$
(C) $T_n < \frac{\pi}{3\sqrt{3}}$ (D) $T_n > \frac{\pi}{3\sqrt{3}}$

10. Let $f(x)$ be a non-constant twice differentiable function defined on $(-\infty, \infty)$ such that $f(x) = f(1-x)$ and $f'\left(\frac{1}{4}\right) = 0$. Then, [IIT-JEE-2008 (Paper-1)]

- (A) $f''(x)$ vanishes at least twice on $[0, 1]$ (B) $f'\left(\frac{1}{2}\right) = 0$
(C) $\int_{-\frac{1}{2}}^{\frac{1}{2}} f\left(x + \frac{1}{2}\right) \sin x dx = 0$ (D) $\int_0^{\frac{1}{2}} f(t) e^{\sin \pi t} dt = \int_{\frac{1}{2}}^1 f(1-t) e^{\sin \pi t} dt$

11. If $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1 + \pi^x) \sin x} dx$, $n = 0, 1, 2, \dots$, then

[IIT-JEE-2009 (Paper-2)]

(A) $I_n = I_{n+2}$

(B) $\sum_{m=1}^{10} I_{2m+1} = 10\pi$

(C) $\sum_{m=1}^{10} I_{2m} = 0$

(D) $I_n = I_{n+1}$

12. The value(s) of $\int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx$ is/are

[IIT-JEE-2010 (Paper-1)]

(A) $\frac{22}{7} - \pi$

(B) $\frac{2}{105}$

(C) 0

(D) $\frac{71}{15} - \frac{3\pi}{2}$

13. Let $f(x) = 7\tan^8 x + 7\tan^6 x - 3\tan^4 x - 3\tan^2 x$ for all $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then the correct expression(s) is(are)

[JEE (Adv)-2015 (Paper-2)]

(A) $\int_0^{\pi/4} xf(x) dx = \frac{1}{12}$

(B) $\int_0^{\pi/4} f(x) dx = 0$

(C) $\int_0^{\pi/4} xf(x) dx = \frac{1}{6}$

(D) $\int_0^{\pi/4} f(x) dx = 1$

14. The option(s) with the values of a and L that satisfy the following equation is(are)

$$\frac{\int_0^{4\pi} e^t (\sin^6 at + \cos^4 at) dt}{\int_0^{\pi} e^t (\sin^6 at + \cos^4 at) dt} = L?$$

[JEE (Adv)-2015 (Paper-2)]

(A) $a = 2, L = \frac{e^{4\pi} - 1}{e^{\pi} - 1}$

(B) $a = 2, L = \frac{e^{4\pi} + 1}{e^{\pi} + 1}$

(C) $a = 4, L = \frac{e^{4\pi} - 1}{e^{\pi} - 1}$

(D) $a = 4, L = \frac{e^{4\pi} + 1}{e^{\pi} + 1}$

15. Let $f'(x) = \frac{192x^3}{2 + \sin^4 \pi x}$ for all $x \in \mathbb{R}$ with $f\left(\frac{1}{2}\right) = 0$. If $m \leq \int_{1/2}^1 f(x) dx \leq M$, then the possible values of m and M are

[JEE (Adv)-2015 (Paper-2)]

(A) $m = 13, M = 24$

(B) $m = \frac{1}{4}, M = \frac{1}{2}$

(C) $m = -11, M = 0$

(D) $m = 1, M = 12$

16. For $a \in \mathbb{R}$, $|a| > 1$, let $\lim_{n \rightarrow \infty} \left(\frac{1 + \sqrt[3]{2} + \dots + \sqrt[3]{n}}{n^{7/3} \left(\frac{1}{(an+1)^2} + \frac{1}{(an+2)^2} + \dots + \frac{1}{(an+n)^2} \right)} \right) = 54$. Then the possible value(s) of a is/

are

(A) 7

(B) -6

(C) 8

(D) -9

[JEE (Adv)-2019 (Paper-2)]

17. Which of the following inequalities is/are TRUE?

[JEE (Adv)-2020 (Paper-1)]

(A) $\int_0^1 x \cos x \, dx \geq \frac{3}{8}$

(B) $\int_0^1 x \sin x \, dx \geq \frac{3}{10}$

(C) $\int_0^1 x^2 \cos x \, dx \geq \frac{1}{2}$

(D) $\int_0^1 x^2 \sin x \, dx \geq \frac{2}{9}$

18. Consider the equation

[JEE (Adv)-2022 (Paper-1)]

$$\int_1^e \frac{(\log_e x)^{1/2}}{x(a - (\log_e x)^{3/2})^2} dx = 1, a \in (-\infty, 0) \cup (1, \infty).$$

Which of the following statements is/are TRUE?

(A) No a satisfies the above equation

(B) An integer a satisfies the above equation

(C) An irrational number a satisfies the above equation

(D) More than one a satisfy the above equation

Linked Comprehension Type Questions

Paragraph for Q.Nos. 19 to 21

Consider the functions defined implicitly by the equation $y^3 - 3y + x = 0$ on various intervals in the real line. If $x \in (-\infty, -2) \cup (2, \infty)$, the equation implicitly defines a unique real valued differentiable function $y = f(x)$. If $x \in (-2, 2)$, the equation implicitly defines a unique real value differentiable function $y = g(x)$ satisfying $g(0) = 0$.

[IIT-JEE-2008 (Paper-1)]

Choose the correct answer :

19. If $f(-10\sqrt{2}) = 2\sqrt{2}$, then $f''(-10\sqrt{2}) =$

(A) $\frac{4\sqrt{2}}{7^3 3^2}$

(B) $-\frac{4\sqrt{2}}{7^3 3^2}$

(C) $\frac{4\sqrt{2}}{7^3 3}$

(D) $-\frac{4\sqrt{2}}{7^3 3}$

20. The area of the region bounded by the curve $y = f(x)$, the x -axis, and the lines $x = a$ and $x = b$, where $-\infty < a < b < -2$, is

(A) $\int_a^b \frac{x}{3[(f(x))^2 - 1]} dx + bf(b) - af(a)$

(B) $-\int_a^b \frac{x}{3[(f(x))^2 - 1]} dx + bf(b) - af(a)$

(C) $\int_a^b \frac{x}{3[(f(x))^2 - 1]} dx - bf(b) + af(a)$

(D) $-\int_a^b \frac{x}{3[(f(x))^2 - 1]} dx - bf(b) + af(a)$

21. $\int_{-1}^1 g'(x) dx =$

- (A) $2g(-1)$ (B) 0
(C) $-2g(1)$ (D) $2g(1)$

Paragraph for Q.Nos. 22 and 23

Given that for each $a \in (0, 1)$, $\lim_{h \rightarrow 0^+} \int_h^{1-h} t^{-a}(1-t)^{a-1} dt$ exists. Let this limit be $g(a)$. In addition, it is given that the function $g(a)$ is differentiable on $(0, 1)$.

[JEE (Adv)-2014 (Paper-2)]

Choose the correct answer :

22. The value of $g\left(\frac{1}{2}\right)$ is

- (A) π (B) 2π
(C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

23. The value of $g'\left(\frac{1}{2}\right)$ is

- (A) $\frac{\pi}{2}$ (B) π
(C) $-\frac{\pi}{2}$ (D) 0

Paragraph for Q.Nos. 24 and 25

Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be a thrice differentiable function. Suppose that $F(1) = 0$, $F(3) = -4$ and $F'(x) < 0$ for all $x \in \left(\frac{1}{2}, 3\right)$.

Let $f(x) = xF(x)$ for all $x \in \mathbb{R}$.

[JEE (Adv)-2015 (Paper-2)]

Choose the correct answer :

24. The correct statement(s) is(are)

- (A) $f'(1) < 0$
(B) $f(2) < 0$
(C) $f'(x) \neq 0$ for any $x \in (1, 3)$
(D) $f'(x) = 0$ for some $x \in (1, 3)$

25. If $\int_1^3 x^2 F'(x) dx = -12$ and $\int_1^3 x^3 F''(x) dx = 40$, then the correct expression(s) is (are)

- (A) $9f(3) + f(1) - 32 = 0$
(B) $\int_1^3 f(x) dx = 12$
(C) $9f(3) - f(1) + 32 = 0$
(D) $\int_1^3 f(x) dx = -12$

Matrix-Match / Matrix Based Type Questions

26. Match the integrals in **Column-I** with the values in **Column-II**.

[IIT-JEE-2007 (Paper-1)]

Column-I

Column-II

(A) $\int_{-1}^1 \frac{dx}{1+x^2}$

(p) $\frac{1}{2} \log\left(\frac{2}{3}\right)$

(B) $\int_0^1 \frac{dx}{\sqrt{1-x^2}}$

(p) $2 \log\left(\frac{2}{3}\right)$

(C) $\int_2^3 \frac{dx}{1-x^2}$

(r) $\frac{\pi}{3}$

(D) $\int_1^2 \frac{dx}{x\sqrt{x^2-1}}$

(s) $\frac{\pi}{2}$

27. Match List-I with List-II.

[JEE (Adv)-2014 (Paper-2)]

List-I

List-II

P. The number of polynomials $f(x)$ with non-negative integer coefficients of degree ≤ 2 , satisfying $f(0) = 0$ and

1. 8

$\int_0^1 f(x)dx = 1$, is

Q. The number of points in the interval

2. 2

$[-\sqrt{13}, \sqrt{13}]$ at which $f(x) = \sin(x^2) + \cos(x^2)$ attains its maximum value, is

R. $\int_{-2}^2 \frac{3x^2}{(1+e^x)} dx$ equals

3. 4

S. $\frac{\left(\int_{-\frac{1}{2}}^{\frac{1}{2}} \cos 2x \log\left(\frac{1+x}{1-x}\right) dx\right)}{\left(\int_0^{\frac{1}{2}} \cos 2x \log\left(\frac{1+x}{1-x}\right) dx\right)}$ equals

4. 0

	P	Q	R	S
(A)	3	2	4	1
(B)	2	3	4	1
(C)	3	2	1	4
(D)	2	3	1	4

Integer / Numerical Value Type Questions

28. For any real number x , let $[x]$ denote the largest integer less than or equal to x . Let f be a real valued function defined on the interval $[-10, 10]$ by $f(x) = \begin{cases} \{x\} & \text{if } [x] \text{ is odd,} \\ 1 - \{x\} & \text{if } [x] \text{ is even} \end{cases}$. Then the value of $\frac{\pi^2}{10} \int_{-10}^{10} f(x) \cos \pi x \, dx$ is

[IIT-JEE-2010 (Paper-1)]

29. The value of $\int_0^1 4x^3 \left\{ \frac{d^2}{dx^2} (1-x^2)^5 \right\} dx$ is

[JEE (Adv)-2014 (Paper-1)]

30. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \begin{cases} [x], & x \leq 2 \\ 0, & x > 2 \end{cases}$, where $[x]$ is the greatest integer less than or equal to x . If $I = \int_{-1}^2 \frac{xf(x^2)}{2+f(x+1)} dx$, then the value of $(4I - 1)$ is

[JEE (Adv)-2015 (Paper-1)]

31. If $\alpha = \int_0^1 (e^{9x+3\tan^{-1}x}) \left(\frac{12+9x^2}{1+x^2} \right) dx$ where $\tan^{-1}x$ takes only principal values, then the value of $\left(\log_e |1+\alpha| - \frac{3\pi}{4} \right)$ is

[JEE (Adv)-2015 (Paper-2)]

32. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous odd function, which vanishes exactly at one point and $f(1) = \frac{1}{2}$. Suppose that

$$F(x) = \int_{-1}^x f(t) dt \text{ for all } x \in [-1, 2] \text{ and } G(x) = \int_{-1}^x t |f(f(t))| dt \text{ for all } x \in [-1, 2]. \text{ If } \lim_{x \rightarrow 1} \frac{F(x)}{G(x)} = \frac{1}{14}, \text{ then the value}$$

of $f\left(\frac{1}{2}\right)$ is

[JEE (Adv)-2015 (Paper-2)]

33. The value of the integral $\int_0^{\frac{1}{2}} \frac{1+\sqrt{3}}{(x+1)^2(1-x)^6} dx$ is _____.

[JEE (Adv)-2018 (Paper-2)]

34. If $I = \frac{2}{\pi} \int_{-\pi/4}^{\pi/4} \frac{dx}{(1+e^{\sin x})(2-\cos 2x)}$; then $27I^2$ equals _____.

[JEE (Adv)-2019 (Paper-1)]

35. The value of the integral $\int_0^{\pi/2} \frac{3\sqrt{\cos \theta}}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^5} d\theta$ equals _____.

[JEE (Adv)-2019 (Paper-2)]

36. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that its derivative f' is continuous and $f(\pi) = -6$. If $F: [0, \pi] \rightarrow \mathbb{R}$

is defined by $F(x) = \int_0^x f(t) dt$, and if

$$\int_0^\pi (f'(x) + F(x)) \cos x \, dx = 2,$$

then the value of $f(0)$ is _____

[JEE (Adv)-2020 (Paper-2)]

Question Stem for Question Nos. 37 and 38

Let $g_i: \left[\frac{\pi}{8}, \frac{3\pi}{8}\right] \rightarrow \mathbb{R}$, $i = 1, 2$, and $f: \left[\frac{\pi}{8}, \frac{3\pi}{8}\right] \rightarrow \mathbb{R}$ be functions such that $g_1(x) = 1$, $g_2(x) = |4x - \pi|$ and $f(x) = \sin^2 x$,

for all $x \in \left[\frac{\pi}{8}, \frac{3\pi}{8}\right]$. Define

[JEE (Adv)-2021 (Paper-2)]

37. The value of $\frac{16S_1}{\pi}$ is _____.

38. The value of $\frac{48S_2}{\pi^2}$ is _____.

39. For any real number x , let $[x]$ denote the largest integer less than or equal to x . If $I = \int_0^{10} \left[\sqrt{\frac{10x}{x+1}} \right] dx$, then the value of $9I$ is _____.

[JEE (Adv)-2021 (Paper-2)]

40. The greatest integer less than or equal to $\int_1^2 \log_2(x^3 + 1) dx + \int_1^{\log_2 9} (2^x - 1)^{\frac{1}{3}} dx$ is _____.

[JEE (Adv)-2022 (Paper-2)]

